higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1 True

1.2 False – thickness is influenced by the type of additives added to the furnish

1.3 True

1.4 False – evaporation increases the solids of the valuable components.

1.5 True

1.6 False – fibrillation takes place during refining.

1.7 False – dandy roll improves web formation by introducing shear in the mix.

1.8 True

1.9 True

1.10 True

(10 × 1) **[10]**

QUESTION 2

2.1 F
2.2 I
2.3 G
2.4 A
2.5 D
2.6 B
2.7 H
2.8 E
2.9 C
2.10 J

(10 × 1) **[10]**

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QUESTION 3

3.1 Box that directs hot dry air onto the wet sheets to dry it

3.2 A process used to make a solid particle behave like a fluid

3.3 Any chemical used to improve the stretching ability of paper when wet or dry

3.4 The amount of energy released by a substance when it is combusted

3.5 The total kilograms of water vaporised in an evaporator per kilogram of steam supplied

(5 × 2) **[10]**

TOTAL SECTION A: 30

SECTION B

QUESTION 4

4.1 4.1.1
- *Low-consistency* pulpers operate at less than 6% and are used to pulp recovered paper for manufacturing packaging paper and board.
- *High-consistency* pulpers operate above 10% and are used for slushing coated grades and printed material like magazines.

(2 × 2) (4)

4.1.2
- Vat
- Rotor
- The drive equipment for driving the rotor (3)

4.1.3
- Drum pulpers
- Horizontal pulpers (2)

4.2
- In contrast to forward cleaners, the accepts are taken from the bottom of the cleaner while the rejects are taken from the top.
- The volume of rejects of the reverse cleaner is much higher than the rejects of the forward cleaner.
- It therefore requires many stages to recover the bulk of the good fibre. (3)

[12]

QUESTION 5

5.1
A – Inlet manifold
B – Headbox
C – Slice
D – Breast roll
E – Rubber deckle
F – Forming board
G – Foils
H – Table rolls
I – Forming fabric or wire
J, K – Vacuum boxes
L – Dandy roll
M – Suction couch roll
N – Lump breaker
O – Guide roll
P – Wire-turning roll/forward-drive roll
Q – Stretch roll
R – Wash roll
S – Stretch roll (18 × ½ (9)

5.2 J – Controls drainage before the dandy roll.
N – Improves water removal at couch and consolidate the sheet.
P – Helps the couch roll to drive the wire (3 x 1) (3)

5.3 It is possible that the sheet may have pin holes or streaks and an additional loss of fines will occur. (2)
[14]

QUESTION 6

6.1 The sheet is brought into contact with the hot surface of the drying cylinders. It is normally done on a multi-cylinder drying system where the contact side of the sheet alternates as it runs along the top- and bottom-dryer cylinders. (4)

6.2 6.2.1 Paper machine operation and control
6.2.2 Check hydraulics or pneumatics
6.2.3 Start-up methods or setting up of instrumentation
6.2.4 Check ride roll alignment or slide movement
(4 × 1) (4)

6.3
- Rider roll relief force profile
- Drum torque differential (or speed differential) profile
- Speed profile
- Tension
- Drum diameter
- Paper grade
- Roll diameter
- Time between paper making and winding
- Set location from parent reel (Any 5 × 1) (5)

[13]

QUESTION 7

7.1
- Large reduction of sulphur losses in the direct contact evaporator
- Reduces corrosion in the multiple effect evaporators
- Reduces the generation of hydrogen sulphide
- Improves combustion in the recovery furnace
- Lowers lime consumption (5 x 1) (5)

7.2 7.2.1 Heating value is the amount of energy that is released by a substance when it is combusted. (1)

7.2.2 Liquor will have a high heating value if it has a high amount of carbon and little oxygen and sodium. (2)

7.3
- Install a baffle plate in the evaporator.
  The baffle plate breaks the bubbles and reduces frothing.
- Use steam jets.
  The jets are directed against the surface of the foam. (Any 1 x 4) (4)

7.4 7.4.1
- Evaporator
- Scrubber
- Reactor (3 x 1) (3)

7.4.2 Stainless steel
Because NSSC spent liquors are very corrosive (2 x 1) (2)
[17]

QUESTION 8

8.1 It converts used sodium compounds into active sodium hydroxide and sodium sulphide. (2)

8.2
- Clarifier
- Slaker and causticisers
- Mud washers
- Storage tank
- Lime kiln
- Gas producer
- Oil burner (7 x 1) (7)

8.3 The process deposits a lot of scale on equipment and the cost associated with cleaning the equipment would be high. There is a possibility of disposing the spent liquor in the sea. (3)

8.4 $S + O_2 \rightarrow SO_2$ (2)
[14]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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